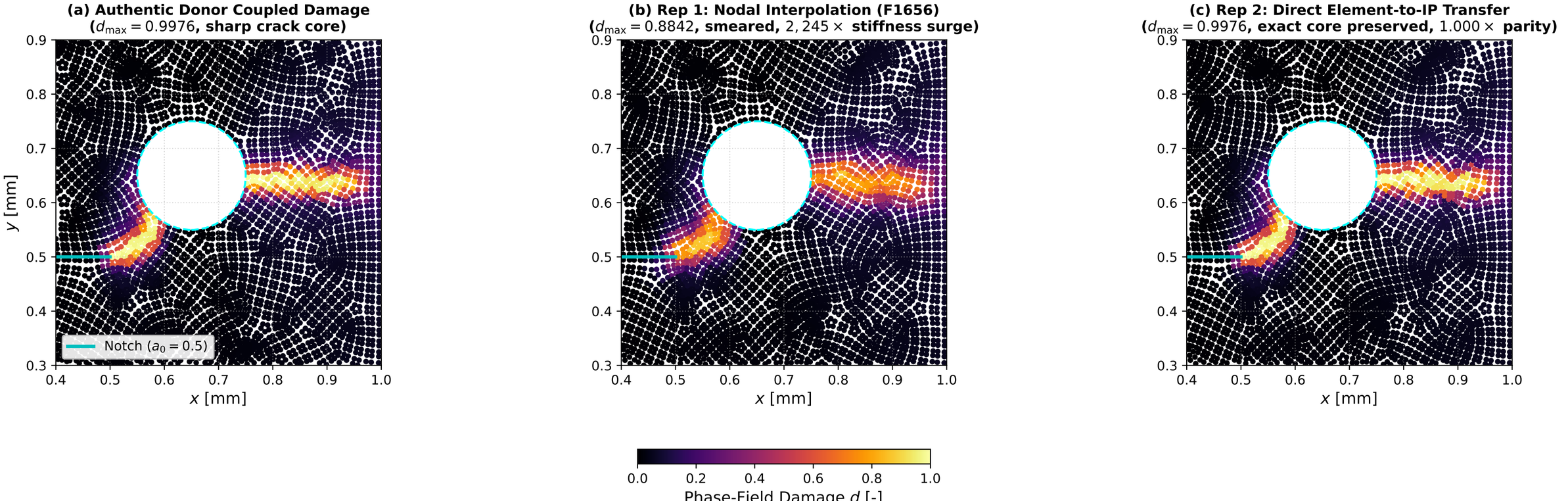

(a) Authentic Donor Coupled Damage
($d_{max} = 0.9976$, sharp crack core)
(b) Rep 1: Nodal Interpolation (F1656)
($d_{max} = 0.8842$, smeared, $2,245\times$ stiffness surge)
(c) Rep 2: Direct Element-to-IP Transfer
($d_{max} = 0.9976$, exact core preserved, $1.000\times$ parity)
Notch ($a_0 = 0.5$)
y [mm]
x [mm]
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
0.0
0.2
0.4
0.6
0.8
1.0
Phase-Field Damage d [-]